

Chapter 38 — Core Principles of Perioperative Management in Children

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PAEDIATRIC ANAESTHESIA

Perioperative Home

- The American Academy of Pediatrics (AAP) defined the **patient/family-centred medical home** over 20 years ago: accessible, family-centred, continuous, comprehensive, culturally sensitive care.
- This led to the **perioperative surgical home (PSH)**, defined by the American Society of Anesthesiologists (ASA) as a patient-centred continuity-of-care model built on **shared decision making**, improving experience and outcomes while **controlling cost**.
- Driver: the **Patient Protection and Affordable Care Act (PPACA) 2010**, which introduced **accountable care organisations (ACOs)** — providers **jointly accountable** for quality targets and cost reduction.
- Care is divided into **preoperative → intraoperative → postoperative → postdischarge** groups; multidisciplinary members of each build **standardised evidence-based pathways**.
- These include **enhanced recovery after surgery (ERAS)**: early mobilisation, **multimodal analgesia with less opioid**, early drain removal, early diet, shorter stay.
- A paediatric PSH led jointly by **surgeons and anaesthesiologists** improves resource use and patient and provider satisfaction.

Anaesthesia and Potential Neurocognitive Effects

- **US Food and Drug Administration (FDA) statement, 14 December 2016**: repeated or lengthy general anaesthesia or sedation in **children under 3 years** or **third-trimester pregnancy** may affect brain development.
- Basis: **animal studies** showing neuroapoptosis and cognitive deficit, extrapolated to the third-trimester fetus, newborn and child up to 3 years.
- Many operations cannot be deferred; for **non-emergent surgery** weigh risk and benefit, especially if **longer than 3 hours** or **multiple procedures** are needed.
- **Smart TOTs**: FDA and International Anesthesia Research Society partnership conducting long-term human studies.

HUMAN STUDIES OF EARLY ANAESTHETIC EXPOSURE

Study	Design	Population and exposure	Verdict
GAS (General Anesthesia compared to Spinal anesthesia; 2016, 2019)	International multicentre randomised controlled trial	Infants under 60 weeks, born after 26 weeks, inguinal hernia repair; awake-regional vs sevoflurane ; anaesthesia under 1 h	No difference in cognitive scores at 2 and 5 years
PANDA (Pediatric Anesthesia NeuroDevelopment Assessment; 2016)	Sibling-matched observational cohort	Healthy children, single exposure under 3; intelligence quotient (IQ), neurocognition and behaviour at 8–15 years	No difference in mean IQ ; prolonged or repeated exposure not assessed
MASK (Mayo Anesthesia Safety in Kids; 2018)	Cohort, 998 children	Anaesthesia before 3; duration not reported	Negligible deficit after single exposure

Study	Design	Population and exposure	Verdict
Large registry comparison	Exposed vs unexposed	Exposure before 4; duration not reported	Slightly lower school grades and IQ

- Pattern: randomised and sibling-matched studies are negative; the only positive signal is observational and lacks exposure duration.
- Bottom line: the warning rests on **preclinical data**; risk from **routine short procedures in early life is negligible, if it exists at all**.
- Stay current; many children's hospitals issue an **institutional statement** for families.
- In non-elective surgery, address each concern: a **single exposure under 1 hour** has not been shown to cause neurocognitive dysfunction, and children deserve **high-quality anaesthesia and analgesia** — inadequate analgesia itself increases morbidity.

KEY POINTS

- The FDA warns that repeated or lengthy (more than 3 hours) general anaesthesia or sedation in children under 3 or in third-trimester pregnancy may affect brain development.
- Inform families of FDA recommendations while reassuring them that human studies show no difference in neurocognitive deficit.

Concept — The FDA warning is animal-derived and framed around repeated or lengthy exposure under age 3; the human landmark studies address single short exposures and are reassuring. Counselling holds both without conflating them.

PREOPERATIVE ASSESSMENT AND PREPARATION FOR SURGERY

Anaesthesia Risk

- **Risk-raising conditions**: upper respiratory infection, asthma exacerbation, dehydration, uncontrolled diabetes, prematurity, febrile illness, comorbidities.
- **Strict adherence to ASA nil per os (NPO) guidelines** reduces aspiration.
- Common risks: postoperative nausea and vomiting, bronchospasm, laryngospasm, stridor, emergence delirium, maladaptive behaviour. Rare and severe: drug allergy, anaphylaxis, malignant hyperthermia, death.
- The anaesthesia team usually counsels, but the **paediatric urologist must be well versed**.

Preoperative Setting

- Preparation is the **foundation** of safety, good outcome and family satisfaction.
- **Pain is shaped by expectation**: give realistic expectations in **age-appropriate language** for a better emotional and physical outcome.
- **Certified child life specialists (CCLSs)** help children and families through anxiety-provoking care; use preoperatively where possible.
- **Information technology devices** (computers, smartphones, handheld games) are accessible, cheap and reduce distress.
- **Virtual reality-guided theatre tours** reduce anxiety and improve compliance at induction.
- **Preoperative anxiety affects up to half** of children and **predicts** higher pain scores, slower recovery after discharge and **emergence delirium** (agitation, crying, thrashing in recovery).

- **New maladaptive behaviours** — anxiety, night crying, enuresis, separation anxiety, tantrums — occur in up to half after surgery; most improve, some persist.

Concept — *Preoperative anxiety is an upstream causal variable: it drives pain, emergence delirium and later behaviour, so reducing child and parental anxiety is a clinical intervention, not a courtesy.*

Special Preoperative Considerations

Jehovah's Witness

- Belief: life resides in blood, and transfusion compromises the soul; parents may refuse on the child's behalf.
- The **physician's legal and ethical obligation is to the child, not the parents.**
- AAP (2013): respect religion and **avoid unnecessary polarisation** — but a child will **not be allowed to die for lack of blood.**
- Authors' institution: **separate consent** that refusal is overridden in life-threatening instances, with an **automatic court order.** Know state law and institutional policy.
- **Blood-sparing optimisation:** correct anaemia and coagulopathy; stop bleeding-promoting drugs and herbal supplements → less morbidity, mortality and transfusion.

Department of Social Services

- Children in custody need consent from a **court-ordered guardian or designee;** obtain it beforehand or ensure availability on the day.

Pain Assessment Tools

- International Association for the Study of Pain (IASP): pain is an **unpleasant sensory and emotional experience** associated with actual or potential tissue damage — harder to assess in children with limited cognition and communication.
- **Self-report is the gold standard,** but is **impossible under 3 years.**
- Common self-report scales: **Wong-Baker FACES** and the **numeric rating scale.**

OBJECTIVE PAIN SCALES BY POPULATION

Population	Scale
Premature infants	PIPP — Premature Infant Pain Profile
Neonates	NIPS — Neonatal Infant Pain Scale; CRIES — crying, oxygenation, vital signs, expression, sleeplessness
Nonverbal and cognitively impaired	FLACC — Face, Legs, Activity, Cry, Consolability; CHEOPS — Children's Hospital of Eastern Ontario Pain Scale

- **FLACC**, the usual choice here, is **confounded by emergence delirium** in the immediate postoperative period.
- Characterise pain beyond intensity: **location, duration, quality, aggravating and relieving factors.** It **may not be at the surgical site,** and cannot be treated until located.

Figure legends

- **Fig. 38.1 Age-based self-assessment pain tools validated in children** (data from Coté and Lerman, *A Practice of Anesthesia for Infants and Children*, 5th ed., 2013): usability rises with age in the order **Pieces of Hurt → FACES → visual analogue → numerical rating scales.**

- **Fig. 38.2 Numeric scale for patients' rating of pain intensity** (Wong-Baker FACES Foundation, 2016): 0–10 scale (no pain, moderate, worst possible) aligned with FACES in steps of two: 0 no hurt, 2 hurts little bit, 4 hurts little more, 6 hurts even more, 8 hurts whole lot, 10 hurts worst.

Concept — The tool follows developmental capacity: self-report where the child can give it, observation where the child cannot — and an observational score measures behaviour, which delirium can imitate.

NPO Status

Fasting Times

- Established by the **ASA (2017)**, **Canadian Anesthesiologists' Society** and **European Society of Anaesthesiology**.

TABLE 38.1 American Society of Anesthesiologists NPO Guidelines

Minimum fast	Ingested material
2 h	Clear liquids
4 h	Breast milk
6 h	Infant formula; nonhuman milk; light meal
8 h	Regular meal

In emergencies, weigh the need for surgery against aspiration risk when NPO status is unknown or unmet. Modified from ASA practice guidelines for preoperative fasting, Anesthesiology 2017.

- The table tracks **gastric emptying**: clear fluid → breast milk → formula, nonhuman milk, light meal → regular meal.
- Trap: **breast milk (4 h) is not formula (6 h)**; nonhuman milk counts as a light meal.

CLEAR-LIQUID FASTING: GUIDELINE BODIES COMPARED

Body	Clear liquids
ASA 2023 modular update	2 h — no evidence found to support 1 h
European guideline (2022)	1 h
Canadian guideline (2022)	1 h

- **Prolonged fasting** causes hunger, thirst, anxiety, nausea, vomiting, pain, hypoglycaemia, electrolyte imbalance, dehydration and **hypotension at induction**.
- Children have **greater potential for fluid loss** than adults.
- **Simple or complex carbohydrate drinks** reduce thirst, hunger and irritability → **allow up to 2 hours** before anaesthesia.
- Outpatients: deficits are minimal when rules are followed, but non-adherence prolongs fasting → **schedule younger children earliest** and educate parents.
- Inpatients: **maintenance intravenous fluid** during the fast prevents dehydration.
- Emergencies: gastric aspiration → **pneumonitis**; **rapid-sequence induction and intubation** protects the full-stomach airway.

Special NPO Situations

Ketogenic diet

- Commonest indication **epilepsy**; explored in autism, amyotrophic lateral sclerosis, multiple sclerosis, Parkinson disease.
- **Stay on the diet**, avoid prolonged fasting; standard NPO (light meal 6 h, clears 2 h).
- Avoid **carbohydrate-containing suspensions**; check every drug for dextrose, carbohydrate or **gluconeogenesis induction** — **mannitol, nitroglycerin, dexamethasone**.

- Consider complete blood count (CBC), electrolytes, albumin, prealbumin; intraoperative glucose and electrolytes if **surgery exceeds 3 hours**.
- Baseline glucose **50–80 mg/dL**; treat only if **below 40 mg/dL or symptomatic**.
- **Blood products** contain variable carbohydrate — avoid where possible.

Enteral tube feeds

- Stop formula **6 hours** before induction; **Pedialyte to 2 hours**.
- Postpyloric feeds carry less reflux and pneumonia risk and some continue them for non-abdominal surgery; the authors apply **standard NPO regardless**, as tube position is uncertain and aspiration morbidity high.

Diabetes and metabolic disorders

- Fasting volume depletion plus withdrawn diabetic drugs → **preoperative hyperglycaemia**.
- Type 1 diabetes affects about **1 in 300** US children.
- **Delay elective surgery until glycaemic control is optimised: no ketonuria**, normal electrolytes, glycated haemoglobin (**HbA1c**) near the age-appropriate range; **first-start case**.
- Admission is not required; some advise it for major surgery.
- **Never induce without a blood glucose**; if surgery is delayed, monitor frequently.

DIABETIC MEDICATION MANAGEMENT

Agent or situation	Instruction	Reason or note
Metformin	Stop 24 h before	Long half-life; lactic acidosis with dehydration, hypoxaemia, poor perfusion
Sulfonylureas, thiazolidinediones	Stop morning of surgery	—
Insulin injections	Normal doses the day before	Authors: half the long-acting dose the night before
Pump — short minor procedure	Continue pump	Plan with anaesthesiologist or endocrinologist
Pump — longer procedure	Intravenous insulin infusion or subcutaneous glargine	Protocol-dependent
Emergency surgery	Intravenous insulin infusion	Frequent glucose checks

- Consider a **preoperative anaesthesia consultation** in metabolic and mitochondrial disorders.
- **Mitochondrial disorders: propofol and lactated Ringer solution** can exacerbate disease → avoided.
- **Glycogen storage and urea cycle disorders** impair glucose metabolism; consider preoperative admission for glucose management.

GLUCOSE STRATEGY WHILE FASTING — THE GOVERNING CONTRAST

Condition	Glucose strategy
Ketogenic diet	Withhold carbohydrate (drugs, suspensions, blood)
Glycogen storage, urea cycle disorders	Cannot tolerate glucose-free periods → glucose-containing fluids throughout the fast
Type 1 diabetes	Glucose measured before induction; insulin adjusted, not stopped

Medication administration

- Medicines may be taken with a clear fluid **2 hours or more** before surgery; consult an anaesthesiologist if unsure.

MORNING-OF-SURGERY MEDICINES

Take	Do not take
Antiseizure; gastro-oesophageal reflux; antihypertensives except angiotensin-converting enzyme (ACE) inhibitors ; chronic pain; behaviour and mood; asthma	Warfarin and aspirin — stop 7–10 days before ; herbal medicines; ACE inhibitors
Never stop abruptly without supervision: drugs with dangerous rebound, e.g. clonidine	

Concept — NPO rules balance aspiration against dehydration and hypoglycaemia; each special situation changes where the glucose and fluid come from, never whether the stomach is protected.

Preoperative Studies**Laboratory Tests**

- **Routine bloods are not indicated** in healthy children for low-risk procedures: low yield, anxiety-provoking, not cost-effective.
- Reserve for **complex procedures** with haemodynamic shifts or major blood loss, and for premature infants, metabolic disorders, personal or family bleeding history, complex conditions.
- Bleeding screens: **HEMSTOP** and the **International Society on Thrombosis and Haemostasis (ISTH) Bleeding Assessment Tool**.
- **Pregnancy testing:** ASA — offer to females of childbearing age and whenever the result alters management; authors test all females of menstrual age with urine human chorionic gonadotrophin (hCG).
- **Trap: false-positive urine hCG after augmentation cystoplasty.**

Chest Radiographs and Electrocardiogram

- **Not routine.** A murmur alone does not warrant an electrocardiogram (ECG); pathological murmurs or cardiopulmonary symptoms → cardiology.
- **Endocarditis prophylaxis** guidance changes often — know and apply the current version.
- Chest radiograph for lung disease or conditions affecting lung function.
- **Screen all adolescents for vaping:** suspected e-cigarette, or vaping, product use-associated lung injury (EVALI) → chest radiograph, computed tomography (CT) and/or pulmonology.

Pulmonary Function Testing

- **No evidence for routine** testing to predict respiratory events; a baseline helps in **severe respiratory disease and severe scoliosis** to allow optimisation.

Urine Evaluation

- Evaluate urinary symptoms before any urological procedure; suspected infection → **urinalysis and culture**.
- When the **urinary tract is opened**, infection risk rises → preoperative culture guides antibiotic choice.
- **Risk factors for a positive intraoperative culture:** neurogenic bladder, prior positive culture, female sex, constipation, urolithiasis.
- **Treat positive cultures before elective surgery.**
- **Asymptomatic bacteriuria matters in endourology** (mucosal injury) → treat with culture-specific antibiotics.

Concept — Preoperative testing is indication-driven, not routine: each test must answer a question that changes management.

COMORBIDITIES: PERIOPERATIVE PLANNING AND POSTOPERATIVE CONSIDERATIONS

Premature Infants

- **Premature:** born before **37 weeks**; **micro-preemie:** before **26 weeks or under 800 g**. Immature organs raise anaesthetic risk.
- **Surfactant deficiency**, especially below **34 weeks** → respiratory distress syndrome; also bronchopulmonary dysplasia (BPD) and chronic lung disease.
- Anaesthesia → low lung volumes and ventilation/perfusion mismatch → hypoxia, oxygen toxicity, barotrauma.
- **Postoperative apnoea** peaks in the first **12–23 hours** and can be fatal.

23-HOUR OBSERVATION FOR POSTOPERATIVE APNOEA

Preterm infant	Term infant
Under 50–60 weeks' postgestational age , with cardiorespiratory monitoring	Under 44 weeks' postgestational age

- **Cardiovascular:** poor diastolic function, low left ventricular compliance; small blood volume → small losses cause hypotension and shock.
- **Retinopathy of prematurity** (especially under 28 weeks) is worsened by hyperoxia → accept saturation in the **low 90s**.
- **Hypothermia:** immature skin, little brown fat, high surface-area ratio → **warm environment mandatory**.
- **Large fluid losses** from skin and wound evaporation and poor urinary concentration.
- Surgery in preterm infants is linked to **reduced white matter on magnetic resonance imaging (MRI)** and lower scores at 2 years → **operate only when urgent or emergent**.

Concept — Every organ system in the preterm infant has less reserve — lungs, heart, blood volume, skin, kidneys, retina and brain — so each hazard is amplified and elective surgery is best deferred.

Asthma and Pulmonary Disease

- Asthma and BPD are the common paediatric lung disorders.
- **Optimise control before elective surgery; continue home medicines on the day.**
- Lung disease raises the risk of bronchospasm, laryngospasm and airway collapse.
- **Inhaled β_2 agonist** improves lung function and reduces respiratory adverse events.
- Perioperative steroids are safe but kept for severe disease (wound infection, poor healing); **stress-dose steroids** if systemic steroids have been needed or inhaled steroids are daily.
- **Induction and emergence are the most dangerous times** (mechanical stimulation) → do not examine or stimulate the patient then.

Cardiovascular Anomalies

- **Congenital heart disease (CHD)**, the commonest congenital anomaly (about **1% of newborns**), carries high perioperative arrest, morbidity and mortality.
- The **Pediatric Perioperative Cardiac Arrest (POCA) Registry** is the best risk-stratification source.
- **Highest risk:** functional single ventricle, pulmonary hypertension, left ventricular outflow tract obstruction, dilated cardiomyopathy.

- Adverse events associated with higher ASA class, emergency, surgery within 30 days, inotropes, preoperative cardiopulmonary resuscitation, kidney injury, mechanical ventilation. Critically evaluate surgical need.

Renal Abnormalities

- Rare in children but a significant cause of morbidity and mortality.
- Under 6 years, **developmental abnormalities** are the commonest cause, then hereditary nephropathies; older children mostly have acquired disease.
- **Chronic kidney disease (CKD) predisposes to acute kidney injury (AKI).**
- **Haematological:** anaemia (less oxygen delivery; may need transfusion); **uraemic platelet dysfunction** → bleeding; uraemia also impairs nutrition and gastric emptying.
- Contrast: **protein-losing nephropathy is hypercoagulable** → thrombosis.
- **Cardiac:** cardiomyopathy, heart failure, pulmonary oedema. Urine output may be normal, polyuric, oliguric or anuric.
- **Electrolytes:** hyponatraemia, hyperkalaemia, hypocalcaemia, hyperphosphataemia, metabolic acidosis; **hyperkalaemia → arrhythmia.**
- Work-up: CBC, chemistry, coagulation; chest radiograph and echocardiogram for effusions if heart or lung disease suspected; **ECG for left ventricular hypertrophy** if hypertensive.
- **Continue antihypertensives and immunosuppressants** unless advised otherwise.
- **Dialysis** preoperatively if indicated; **avoid surgery for 4–6 hours after haemodialysis.**
- Analgesia: **regional blocks and epidurals** if no coagulopathy.

ANALGESIA BY CKD STAGE

Stage	Renal function	Analgesia
1	Normal	Nonsteroidal anti-inflammatory drugs (NSAIDs), opioids, tramadol
2	Mild impairment; drugs still excreted	No dose adjustment; watch for deterioration and NSAID effect
3–4	Moderate to severe impairment	Avoid NSAIDs; dose by glomerular filtration rate (GFR), interval by stage; watch sedation

Haematological Disorders

- Haemoglobinopathies are commonly inherited: **α - and β -thalassaemia** and **structural variants (HbS, sickle cell disease)**; also von Willebrand disease, factor deficiencies, platelet disorders.
- Take a **thorough family and personal bleeding history**; **positive family history of excessive bleeding → haematology.**
- **Sickle cell disease:** haemolysis and vaso-occlusion from irreversible sickling; chronic anaemia, pain crisis, acute chest syndrome, stroke, splenic sequestration, cardiomyopathy, heart failure, renal disease.
- Assess end-organ damage; surgeon–anaesthesiologist–haematologist planning; **preoperative transfusion is controversial**; hydroxyurea may help some.
- **Acute chest syndrome is the main cause of death** under anaesthesia and may occur postoperatively.
- **Most important single factor: avoid dehydration**, especially during NPO → **first case**, intravenous fluid during the fast.
- Intraoperative: **normothermia, hydration, analgesia, no hypoxia or acidosis; avoid tourniquets.**

- Postoperative: intravenous fluid until drinking; avoid hypoxia, hypothermia, acidosis, infection; analgesia; **early ambulation**.

Concept — Sickle cell management is trigger avoidance: dehydration, hypoxia, acidosis, cold and stasis each promote sickling, and every perioperative measure removes one of them.

Oncological Disorders

Chemotherapy and Perioperative Considerations

- Toxicity affects heart, liver, kidneys, gut, lungs, marrow and nervous system.
- Surgery may fall at any point in treatment → take a **detailed history of the regimen**, side effects and radiation; testing depends on regimen, symptoms and timing.

CHEMOTHERAPY TOXICITIES RELEVANT TO ANAESTHESIA

Agent or class	Toxicity	Implication
Bleomycin	Dose-dependent interstitial pneumonitis → fibrosis; hypersensitivity pneumonitis, acute chest pain syndrome, pulmonary veno-occlusive disease, bronchiolitis obliterans	Oxygen restriction may be unnecessary; fluid balance better predicts pulmonary morbidity
Cardiotoxic agents (common)	Arrhythmia, blood pressure lability, cardiomyopathy, heart failure, myocarditis, fibrosis	Cardiac assessment
Nephrotoxic agents	Impaired renal function	Renal assessment
Myelosuppressive agents	Pancytopenia	Poor oxygenation, haemorrhage, sepsis
Tyrosine kinase inhibitors	Variable by agent	Agent-specific review

Pheochromocytoma

- Rare **adrenal medullary** neuroendocrine tumours storing noradrenaline and adrenaline.
- Manipulation or induction → **catecholamine storm**: hypertensive crisis, arrhythmia, myocardial ischaemia, pulmonary oedema, stroke.
- Pathophysiology: **low blood volume with hypertension**; high catecholamines → cardiomyopathy. Blood pressure control and fluid support are essential.
- Target blood pressure **below the 50th percentile for age and height**, with an **α-adrenergic antagonist before β-blockade** to avoid crisis from unopposed α stimulation.
- **Admission for 24 to 26 hours** to optimise hydration and blood pressure.
- Postoperative **hypoglycaemia**: catecholamine fall → rebound hyperinsulinaemia and peripheral glucose uptake → **monitor glucose**.

Concept — Pheochromocytoma care follows catecholamine physiology: α-blockade before β-blockade, volume before induction, and glucose surveillance once the catecholamine source is removed.

Spina Bifida

- Neurological, anatomical and comorbid challenges all shape planning.
- **80–90%** have or develop **hydrocephalus**; the **ventriculoperitoneal (VP) shunt is the gold standard**, or endoscopic third ventriculostomy.
- Shunt function must be adequate; **suspected malfunction → reschedule elective surgery**.
- **Scoliosis in about half** reduces thoracic capacity; lung disease may be silent → preoperative respiratory testing helps.

- **Overweight and obesity** are common → metabolic syndrome: diabetes, dyslipidaemia, hypertension, cardiovascular disease, sleep apnoea, osteoporosis, poor skin integrity.
- Theatre: know the skeletal anatomy, **pad pressure points**, position carefully; air mattress for severe deformity with long admission.
- **Autonomic dysreflexia**: lesions **at T6 and above**; noxious stimuli or **bladder distension** trigger dysregulated sympathetic discharge → uncontrolled hypertension, bradycardia, stroke, death.
- **Anaphylactic latex allergy**: bronchoconstriction, vasodilatation and hypotension, capillary leak, coronary vasoconstriction, myocardial depression; fatal without immediate treatment.
- Analgesia is difficult, especially with shunted hydrocephalus: **epidural catheters are safe and effective** in selected patients; **peripheral regional catheters reduce opioid use without raising pain scores** after major surgery; alternatives patient-controlled analgesia (PCA) and intravenous drugs.

Congenital Adrenal Hyperplasia

- Congenital adrenal hyperplasia (CAH): inherited inability to make cortisol; **90–99% from 21-hydroxylase deficiency (CYP21A2 mutations)**.
- Cortisol deficiency ± aldosterone deficiency, plus androgen excess; **salt-losing vs non-salt-losing**.
- Salt-wasting: renal hyponatraemia, hyperkalaemia, acidosis, shock if untreated. Cortisol deficiency → poor contractility, low vascular tone.
- Classic CAH **cannot mount a stress cortisol response** → **stress-dose steroids, preferably intravenous with intravenous fluids**; dose differs for minor and major surgery.
- Adrenaline and cortisol deficiency → **fasting hypoglycaemia** → glucose-containing fluids and monitoring.

KEY POINTS

- Anaesthesia risks include postoperative nausea and vomiting, bronchospasm, laryngospasm, stridor, emergence delirium and maladaptive behaviours.
- NPO violation significantly increases the risk of gastric aspiration and pneumonitis.
- Children are at increased risk of dehydration, so families need NPO education.
- Premature infants have significantly increased perioperative risk.
- Spina bifida patients have multiple comorbidities to consider perioperatively.
- Autonomic dysreflexia is life threatening; paediatric urologists should know its signs and treatment.

Acute Preoperative Events

Sepsis

- A leading cause of paediatric morbidity, mortality and cost; mortality varies widely with severity, risk factors and geography.
- Most deaths are from **shock or multiple organ dysfunction within 48–72 hours** → early recognition and aggressive resuscitation.
- Infection, especially **urosepsis**, is a common cause.
- **Empirical antimicrobials within 1 hour of recognising septic shock**, chosen by history and site.
- Rapid aggressive fluids ± vasopressors; airway support; oxygen titrated to **continuous central venous oxygen saturation (ScvO₂)** in the first hours.
- If surgery is needed: crossmatch, adequate vascular access, chemistry, CBC, coagulation; **septic shock** → **intensive care postoperatively**.

REGIONAL TECHNIQUES IN SHOCK

Peripheral nerve blocks	Spinal and epidural anaesthesia
Effective ; avoid systemic opioid effects in an unstable patient	Generally avoided — haemodynamic effects, possible coagulopathy

Trauma

- Trauma kills more children aged **1–18** than all other causes combined.
- In high-impact blunt trauma the **kidney is injured more often than spleen, liver, pancreas or bowel**; renal injury is usually conservative, but other injuries may force exploration.
- Focused history and examination; **imaging must not delay lifesaving intervention**; focused abdominal sonography for trauma (FAST) may help.
- Bloods can be drawn in theatre. **Trauma panel**: CBC, metabolic panel, coagulation, amylase, lipase, type and screen, urinalysis, β -hCG in females of reproductive age.
- **Always a full stomach; rapid-sequence induction is the gold standard.**
- **Keep theatre warm**: hypothermia → vasoconstriction, low flow, acidosis, consumptive coagulopathy, worse outcomes.
- Untreated pain has lasting psychological and physiological effects: acetaminophen (oral, rectal, intravenous), ibuprofen; **strong opioids for severe pain** unless contraindicated.

Upper Respiratory Tract Infection

- Upper respiratory tract infection (URTI) is the **commonest non-preventable cause of cancellation**; the **COLDS score** grades its severity.
- A prospective study of children 1–5 years having ilioinguinal surgery with a laryngeal mask airway: respiratory adverse events stayed frequent; risk factors were **passive smoking** and **severe URTI in the preceding 15 days**.
- Other independent risk factors: **endotracheal tube, age under 5, prematurity, asthma, airway surgery**.
- Inflammation leaves a **hyperreactive airway for 6–8 weeks** → laryngospasm, stridor, bronchospasm, cough, breath-holding, desaturation, bradycardia, atelectasis, pneumonia.
- **Laryngeal mask airway (LMA) causes fewer complications than an endotracheal tube.**

TIMING OF SURGERY AFTER URTI

Study protocol	Authors' institution
Severe URTI delayed 15 days — shown insufficient	Postpone at least 4 weeks, preferably 6 , to regain respiratory baseline

- Surgeon and anaesthesiologist decide whether to postpone or proceed; if proceeding → **experienced paediatric anaesthesia team**.
- **COVID-19 positive: delay all elective surgery**; emergencies in a **negative-pressure room**.

Urinary Tract Infection

- A best-practice statement advises urinalysis, microscopy and culture before any urological procedure when urinary tract infection (UTI) is suspected.
- **American Urological Association (AUA) guidelines (2018): eradicate infection before urological manipulation** with a culture-directed course.
- **Unrecognised bacteriuria** → sepsis and surgical site infection (SSI). Risk factors for a positive culture are as under Urine Evaluation.
- UTI or bacteraemia → antibiotics and **reschedule until urine is sterile**.

- Where sterilisation is impossible (indwelling catheter, infected stones, emergency) → **perioperative culture** guides treatment if infection follows.
- **Neurogenic bladder: always obtain a preoperative culture** and treat by sensitivity; paediatric guidelines are still needed.

Nonspecific Fever

- Usually **infective**, typically URTI — **viral in about 95%**; allergy and rhinitis contribute.
- **Post-transfusion fever up to 1 week**; also drug allergy, autoimmune disease, malignancy, other comorbidities.
- Surgeon and anaesthesiologist decide whether to cancel; **physical examination is central**.

KEY POINTS

- Antibiotics are the mainstay of sepsis treatment and should be given as soon as possible; obtain cultures if possible.
- Obtain urine cultures preoperatively when indicated and treat positives according to sensitivities.

Concept — *Acute events divide into two decisions: in sepsis and trauma surgery cannot wait, so physiology is supported around it; in URTI, UTI and fever surgery can wait, so the question is how long the risk persists.*

ERAS PROTOCOLS

- ERAS is patient-centred multimodal care that optimises recovery and **preserves the preoperative physical and metabolic state**.
- Paediatric elements: **enhanced counselling, shorter starvation, judicious intravenous fluid, no routine tubes or drains, adjunctive analgesia**.
- Paediatric urology studies **all report at least one benefit** — shorter operative time and stay, earlier feeding and bowel function — with **no rise in complications** (some fewer) or **readmissions**.
- Poor analgesia prolongs stay; **regional anaesthesia** adds theatre time but improves pain and shortens stay.
- Barriers: cultural, racial and health-system variation, lack of consensus, need for administrative support → impaired standardisation and evaluation.

Concept — *Every ERAS element minimises disturbance of the child's baseline — less starvation, less fluid excess, fewer tubes, fewer opioids — and recovery follows from what is not disturbed.*

INTRAOPERATIVE CONSIDERATIONS

Preoperative Antibiotics

- SSIs cause morbidity, mortality, readmission and prolonged stay.
- Principle (AUA 2018): prophylaxis is justified by host defence, likelihood of bacterial invasion and morbidity of infection — **benefit must outweigh antibiotic risk**.
- Paediatric practice is **extrapolated from adults**; AUA gives procedure-specific advice; paediatric urology data are limited and practice is changing.
- Timing for adequate tissue levels: **within 60 minutes of incision; vancomycin and fluoroquinolones within 120 minutes** (long infusion).

TABLE 38.2 Common Antimicrobials Used for Surgical Prophylaxis

Class	Agent	Paediatric dose	Redose (h)
Penicillins	Ampicillin; ampicillin-sulbactam (ampicillin component)	50 mg/kg	2
Penicillins	Piperacillin-tazobactam	2–9 months 80 mg/kg; over 9 months and up to 40 kg 100 mg/kg (piperacillin)	2
Cephalosporins	Cefazolin	30 mg/kg	4
Cephalosporins	Cefuroxime; cefotaxime	50 mg/kg	4; 3
Cephalosporins	Cefoxitin; cefotetan	40 mg/kg	2; 6
Cephalosporins	Ceftriaxone	50–75 mg/kg	—
Monobactam	Aztreonam	30 mg/kg	4
Carbapenem	Ertapenem	15 mg/kg	—
Fluoroquinolones	Ciprofloxacin, levofloxacin, moxifloxacin	10 mg/kg	—
Aminoglycoside	Gentamicin	2.5 mg/kg	—
Glycopeptide	Vancomycin	15 mg/kg	—
Lincosamide	Clindamycin	10 mg/kg	6
Nitroimidazole	Metronidazole	15 mg/kg; neonates under 1200 g 7.5 mg/kg	—
Azole antifungal	Fluconazole	6 mg/kg	—

— = no intraoperative redosing interval. Side effects: allergy including anaphylaxis, gastrointestinal upset, *Clostridium difficile* infection, resistance; renal or hepatic insufficiency may need dose adjustment. Modified from Bratzler et al., Clinical practice guidelines for antimicrobial prophylaxis in surgery (American Society of Health-System Pharmacists, Infectious Diseases Society of America, Surgical Infection Society, Society for Healthcare Epidemiology of America), *Surg Infect* 2013.

- Pattern: **short half-life β -lactams redose every 2 hours**; cefazolin, cefuroxime and aztreonam every 4; long half-life agents (ceftriaxone, ertapenem, vancomycin, gentamicin, fluoroquinolones) are not redosed.
- Traps: **cefazolin 30 mg/kg every 4 hours**; vancomycin and fluoroquinolones start earlier (120 minutes), not later.

Intraoperative and Perioperative Fluid Management

- Both **restrictive and liberal** regimens cause complications → tailor to deficit, maintenance and losses; deficit reflects NPO time and any **bowel preparation**.
- **4-2-1 rule (Holliday and Segar, 1957)**: 4 mL/kg/h for the first 10 kg, 2 mL/kg/h for the next 10 kg, 1 mL/kg/h per kg above 20 kg; 22 kg → 62 mL/h.
- No definitive crystalloid or colloid preference; some argue **lactated Ringer solution** beats normal saline (NS), which can cause **dilutional acidosis**.
- Caveats: **dextrose-containing maintenance under 2–4 months** (immature liver); **no lactate in mitochondrial disease**; **no potassium in dialysed patients**.
- Continue maintenance until oral intake is adequate.

POSTOPERATIVE MAINTENANCE FLUID

Historical	Current
Hypotonic, e.g. 0.45% NS	Isotonic — 0.9% NS or lactated Ringer — with dextrose

- Hyponatraemia (**below 135 mmol/L**) is uncommon, but postoperative risk is higher.

- Mechanism: **stress raises antidiuretic hormone (ADH)** — made in the hypothalamus, released from the posterior pituitary, inserting **aquaporin-2** in the collecting duct → water reabsorption. Normally osmolality-driven, but **nausea, pain and hypoxia** also release it.
- Potassium in NS: **10 mEq/L under 10 kg; 20 mEq/L over 10 kg**.
- Premature kidneys cannot handle large solute loads → tailor maintenance.

Anaphylaxis

- **Commoner in adults than children**; mainly **within minutes of induction**; latex now a less common cause.
- Presentation: cardiovascular collapse — tachycardia, hypotension, severe bronchospasm, hives.
- Management: stop the culprit, 100% oxygen, Trendelenburg, aggressive fluids, albuterol, adrenaline.

Malignant Hyperthermia

- Malignant hyperthermia (MH): rare, potentially fatal hypermetabolic disorder.
- **Early signs**: unexplained hypercarbia, sinus tachycardia, masseter spasm, hyperthermia. **Late signs**: severe metabolic acidosis, peaked T waves (hyperkalaemia), generalised rigidity, rhabdomyolysis.
- Susceptible patient → **first-start case**; **activated charcoal filters** cut machine preparation from 20–104 minutes to **90 seconds**.

Suspected event: stop triggers → call for backup → obtain dantrolene → disconnect until a clean machine replaces the old → arterial line and Foley catheter → MH hotline (800-644-9737) → treat hyperkalaemia and arrest aggressively → admit **24–48 hours** (recrudescence) → report to the MH registry (Malignant Hyperthermia Association of the United States, MHAUS).

- Use a **cognitive aid**. Mortality has fallen to about **5%**, but morbidity remains significant → rapid recognition, aggressive treatment.

ANAPHYLAXIS VS MALIGNANT HYPERTHERMIA

	Anaphylaxis	Malignant hyperthermia
Nature	Immune reaction	Hypermetabolic muscle disorder
Triggers	Neuromuscular blockers, β-lactams, latex	Succinylcholine, volatile agents (not nitrous oxide)
First sign	Collapse within minutes of induction	Unexplained hypercarbia
Specific treatment	Adrenaline	Dantrolene
Marker and follow-up	Tryptase during the event ; postpone surgery	Admit 24–48 h; report to registry

Concept — Both are drug-triggered intraoperative crises, but anaphylaxis is immune and immediate (tryptase, adrenaline) whereas malignant hyperthermia is a muscle hypermetabolic state that announces itself through carbon dioxide first (dantrolene).

LAPAROSCOPIC PROCEDURES

- Introduced to paediatric urology in **1976** as **diagnostic laparoscopy for nonpalpable testes**; use has grown; well tolerated, but paediatric physiology differs.
- **Respiratory**: children have **high chest wall compliance and low functional residual capacity (FRC)**; insufflation elevates the diaphragm → lower FRC and lung compliance, higher airway resistance, ventilation/perfusion mismatch and dead space.
- **Cardiovascular**: little effect with a normal heart, but **bradyarrhythmias** (bradycardia, atrioventricular dissociation, nodal rhythm, asystole) at port placement and insufflation.

- **Renal:** no proven harm to renal or cerebral perfusion, but pneumoperitoneum compresses parenchyma and vessels → higher renovascular resistance, lower renal vein flow → **transient oliguria**; with impaired renal function give **generous volume before and during** pressure rises.
- **Positioning:** pad pressure points; **axillary roll in lateral decubitus** (brachial plexus); neutral neck; dependent leg flexed with a cushion between the knees.
- **Analgesia:** multimodal (NSAIDs plus opioids) speeds recovery; consider regional or neuraxial blocks; pneumoperitoneum causes diffuse abdominal and **shoulder pain**.
- **Carbon dioxide (CO₂) embolism:** large-volume CO₂ enters veins during insufflation → right ventricular outflow obstruction → sudden desaturation and collapse.
- Classic: **falling end-tidal CO₂ with a mill-wheel murmur**. Treatment: **left lateral decubitus (right side up) and Trendelenburg**; aspirate via a central line if present.

Robotic Procedures

- **Robotic pyeloplasty** is the commonest; others: heminephrectomy, ureteroureterostomy, ureteral reimplantation, urachal cyst excision, bladder diverticulectomy, bladder reconstruction.
- All laparoscopic considerations apply.
- **Robot location:** often in shared adult theatres, away from paediatric experts, equipment and monitors → **routine simulation**; **two anaesthesia physicians at induction** for high-risk patients.

Intraoperative Pain Management

- Agree the plan with the anaesthesiologist; it depends on patient, surgery and anaesthesiologist comfort.
- **Epidural analgesia** suits open surgery with large incisions; safe from **2 kg** in experienced hands.
- **Regional and neuraxial blocks under general anaesthesia** are safe and standard.
- Peripheral: penile ring, dorsal penile nerve blocks. Neuraxial: spinal, caudal, site-specific epidural.

CAUDAL VS PUDENDAL BLOCK

Caudal block	Ultrasound-guided pudendal block
No overall consensus on superiority in paediatric urology	For hypospadias repair : shorter stay, lower pain, less opioid, greater satisfaction

- **Spinal anaesthesia** for infants in short procedures (**under 90 minutes**): growing interest, but favoured by only a minority of paediatric urologists.
- Regional and neuraxial analgesia reduce intraoperative opioid use.
- **Multimodal:** preoperative acetaminophen, a cyclo-oxygenase-2 (COX-2) inhibitor and gabapentin in selected patients; **intravenous acetaminophen preferred over rectal** (unpredictable absorption).
- **Intravenous lidocaine:** beneficial, sometimes continued 24 hours; in most ERAS protocols; **count it in the total local anaesthetic dose**.
- **Ketamine and dexmedetomidine infusions** reduce opioid need; **dexmedetomidine is neuroprotective in animal models**.
- **Intravenous ketorolac:** used safely under 6 months, but usually reserved for **6 months and older with normal renal function**.

KEY POINTS

- Antibiotics are the mainstay of SSI prevention; choose by site and likely source of infection.

- Multimodal pain management should be a mainstay of treatment.

Concept — *Intraoperative analgesia is opioid-sparing by construction: every block and adjunct is chosen for the opioid it replaces.*

POSTOPERATIVE MANAGEMENT

Immediate Postoperative Management

- Focus on **respiratory and cardiovascular** status.
- Respiratory events: hypoxia, obstruction, laryngospasm, bronchospasm, post-obstructive pulmonary oedema, aspiration — more frequent if **intubated, ASA 3–4, under 1 year**, respiratory disease, prematurity, current or recent upper respiratory infection.
- **Multiple intubation attempts** → airway oedema → stridor.

Laryngospasm: positive-pressure ventilation → propofol to relax the cords → reintubation if these fail.

Bronchospasm: albuterol/ipratropium nebulisers → if severe, adrenaline, steroids, magnesium, continuous albuterol; admit if needed.

- **CHD** raises cardiac event risk. Postoperative arrest in non-cardiac children is rare — **fewer than 1 per 100,000** anaesthetics; most occur **under 5 years** and **most are preventable** → constant vigilance in the postanaesthesia care unit (PACU).

Emergence delirium

- Especially at **2–8 years**: restlessness, agitation, combativeness, inconsolability; incidence varies widely.
- Usually **within 30 minutes and self-limited**, but distressing and dangerous — self-injury, dislodged catheters and lines, wound damage.
- Exclude pain and normal behaviour (tantrum); the **Pediatric Anesthesia Emergence Delirium (PAED) scale** helps.
- **High preoperative anxiety** raises risk.
- Treatment: **low-dose propofol**, watching the airway; dexmedetomidine, fentanyl, clonidine. Warn parents preoperatively; **treat pre-emptively** in high-risk children.

EMERGENCE DELIRIUM VS PAIN

Emergence delirium	Pain
Child does not recognise parents or surroundings ; inconsolable	Child recognises parents; localises complaint; responds to analgesia

Postanaesthesia Care Unit and Pain Management

- The anaesthesiologist **stays until a pain plan is in place**; titrate opioids and non-opioids to the institutional scale.
- **PCA** works as well as in adults; safe from **4 years if cognitively appropriate**.
- **PCA by proxy** (parent or nurse): set maximum doses, assess pain properly, monitor for respiratory depression and sedation.
- Confirm regional or neuraxial blocks work; **no PACU discharge until pain is controlled**.
- Discharge at baseline function with normal haemodynamics, using the **Aldrete score** and **Post-Anesthetic Discharge Scoring System (PADSS)**.
- Postanaesthesia note (respiratory, cardiovascular, mental status, temperature, pain, nausea, vomiting, hydration) required by the Centers for Medicare & Medicaid Services **within 48 hours**.

Inpatient and Outpatient Postoperative Pain Management

- Most surgical patients have postoperative pain, often **moderate to severe**; in children it is often **undertreated** (hard to assess, fear of opioids).
- Poor control → complications, slower recovery, morbidity, mortality, anxiety and fear.
- Discuss expected pain **preoperatively**.
- **Multimodal: weight-based acetaminophen plus NSAID is the base** and substantially reduces opioid use.
- **Ketorolac** is routine; a Cochrane review found low-quality evidence on its safety and efficacy; **bleeding** remains a concern.
- **Intranasal, rectal and transdermal** routes help where venous access is difficult.
- Opioids for breakthrough pain; side effects nausea, pruritus, constipation, respiratory depression.
- **Codeine**: regulators warn against use in children; **absolutely contraindicated after tonsillectomy for obstructive sleep apnoea**.
- Start with a **scheduled regimen**; as-needed dosing is inadequate.

TABLE 38.3 Pediatric Pain Medication Dosages and Intervals — opioids

Drug	Oral	Intravenous	Frequency	PCA
Morphine	0.15–0.3 mg/kg	0.05–0.1 mg/kg; neonate 0.025–0.05	3–4-hourly; neonate 6-hourly	0.01–0.02 mg/kg every 10 min ± basal 0.01–0.02 mg/kg/h
Hydromorphone	0.04 mg/kg	0.01 mg/kg	3–4-hourly	0.004 mg/kg every 10 min ± basal 0.002–0.004 mg/kg/h
Fentanyl	—	0.5–1 mcg/kg; intranasal 1–2 mcg/kg	30–60-minutely	0.5–1 mcg/kg every 6 min ± basal 0.5–1 mcg/kg/h
Oxycodone	0.1 mg/kg	—	4-hourly	—
Codeine	Not recommended			
Methadone	0.1–0.2 mg/kg (max 5 mg)	0.1 mg/kg (max 2–4 mg)	12–24-hourly	—

TABLE 38.3 Pediatric Pain Medication Dosages and Intervals — non-opioids

Drug	Dose	Frequency	Limits and notes
Acetaminophen	Oral 15 mg/kg; intravenous 15 mg/kg from 2 years	6-hourly	Intravenous FDA-approved from 2 years; lower doses used under 2 — consult paediatric anaesthesia
Ibuprofen	Oral 10 mg/kg	8-hourly	Not under 6 months
Ketorolac	Intravenous 0.5 mg/kg (max 30 mg); loading 1 mg/kg (max 30 mg) once	8-hourly	Limit to 72–96 h (renal toxicity); not under 6 months
Celecoxib	Oral 2–4 mg/kg	12-hourly	Max 200 mg twice daily; FDA-approved only for juvenile idiopathic/rheumatoid arthritis, 2 years and 10 kg or over
Naproxen	Oral 5 mg/kg	12-hourly	Max 500 mg twice daily; 2 years and over only
Gabapentin	Start 2 mg/kg at night → 5 mg/kg/day, titrate to max 35 mg/kg/day, three times daily	8-hourly	Neuropathic pain; drowsiness — titrate every 2–3 days
Pregabalin	1.5 mg/kg/day (max 6 mg/kg/day)	12-hourly	Neuropathic pain after gabapentin failure

Fentanyl: short half-life — for brief acute or procedural pain, not sustained pain. Codeine: black box warning — poor metabolisers get no analgesia; ultrarapid metabolisers convert a lethal amount to morphine. Methadone: long half-

life, accumulates → respiratory depression, death, and cardiovascular events in prolonged QT syndrome — ECG required; consider pain consultation first.

- Pattern: PCA boluses are small fractions of the intravenous bolus with short lockouts; NSAIDs are **age-limited** (not under 6 months; naproxen and celecoxib from 2 years) and ketorolac is **duration-limited** by renal toxicity.
- Traps: **codeine** is genotype-dependent — useless or lethal; **methadone needs an ECG**; intravenous acetaminophen approval starts at 2 years.

OPIOID USE IN CHILDREN

- Overdose and misuse risk is high in children, adolescents and young adults: **access to unused medicines** and developing **reward and habit-formation centres**.
- **No standard paediatric urology prescribing guidelines** → overprescription and leftover household opioids.
- Yet **untreated pain** has lasting psychological and physiological consequences.
- **Prescribing errors** above the maximum daily dose occur, concentrated in the youngest → check dosing carefully.
- Major surgery needing opioids: **limited discharge supply, e.g. 3–5 days**.
- Teach dose, frequency, **storage and disposal**: poor storage and undisposed excess are the **main sources of accidental poisoning in young children and recreational overdose in adolescents**; disposal guidance on the FDA website.

KEY POINTS

- Perioperative complications most often occur in the immediate postoperative period.
- Team members should know the potential postoperative complications and monitor for them.
- Postoperative pain control is important for optimal outcomes; a multimodal drug approach is recommended.
- Proper administration and disposal of unused narcotics prevent overdose.

Concept — *The same drug must be given adequately in hospital and removed from the household after discharge: undertreatment and leftover supply are the twin failures of paediatric opioid practice.*